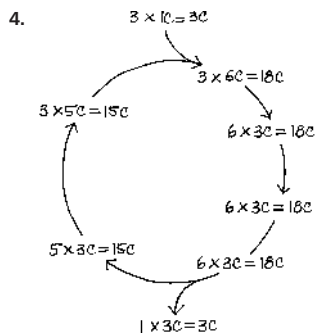




For every three turns of the Calvin cycle, the total number of carbon atoms remains constant because three carbons enter as part of CO_2 molecules, replacing the three that left as part of a G3P molecule. 5. In glycolysis, G3P acts as an intermediate. The six-carbon sugar fructose 1,6-bisphosphate is cleaved into two three-carbon sugars, one of which is G3P. The other is an isomer called dihydroxyacetone phosphate (DHAP), which can be converted to G3P by an isomerase. Because G3P is the substrate for the next enzyme, it is constantly removed, and the reaction equilibrium is pulled in the direction of conversion of DHAP to more G3P. In the Calvin cycle, G3P acts as both an intermediate and a

product. For every three CO_2 molecules that enter the cycle, six G3P molecules are formed, five of which must remain in the cycle and become rearranged to regenerate three five-carbon RuBP molecules. The one remaining G3P is a product, which can be thought of as the result of “reducing” the three CO_2 molecules that entered the cycle into a three-carbon sugar that can later be used to generate energy.

Concept Check 10.5

1. Photorespiration decreases photosynthetic output by adding O_2 , instead of CO_2 , to the Calvin cycle. As a result, no sugar is generated (no carbon is fixed), and O_2 is used rather than generated.
2. Without PS II, no O_2 is generated in bundle-sheath cells. This avoids the problem of O_2 competing with CO_2 for binding to rubisco in these cells.
3. Both problems are caused by a drastic change in Earth's atmosphere due to burning of fossil fuels. The increase in CO_2 concentration affects ocean chemistry by decreasing pH, thus affecting calcification by marine organisms. On land, CO_2 concentration and air temperature are conditions that plants have become adapted to, and changes in these characteristics have a strong effect on photosynthesis by plants. Thus, alteration of these two fundamental factors could have critical effects on organisms all around the planet, in all different habitats.
4. You would expect that C_4 and CAM species would replace many of the C_3 species.

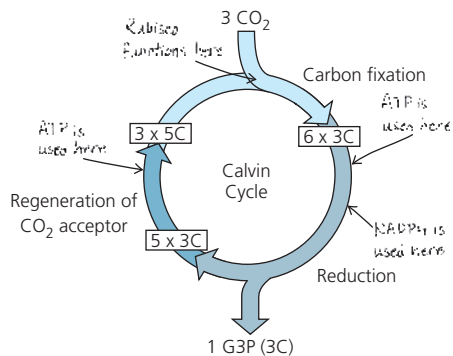
Concept Check 10.6

1. Plants can break down the sugar they make (in the form of glucose) by cellular respiration, producing ATPs for various cellular processes such as endergonic chemical reactions, transport of substances across membranes, and movement of molecules in the cell. ATPs are also used for the movement of chloroplasts during cellular streaming in some plant cells (see Figure 6.26).

Summary of Key Concepts Questions

10.1 Photosynthesizing autotrophs are called producers because they use sunlight as energy to make their own organic molecules (including sugars) from CO_2 and H_2O . Heterotrophs are called consumers because they cannot make their own food and therefore must feed on other organisms, either autotrophs or other heterotrophs. Heterotrophs that consume the remains of dead organisms are called decomposers. Photosynthesis enables almost all organisms to survive because it makes food for photosynthesizing autotrophs, which in turn are food for most heterotrophs. **10.2** CO_2 and H_2O are the products of cellular respiration; they are the reactants in photosynthesis. In respiration, glucose is oxidized to CO_2 and electrons are passed through an electron transfer chain from glucose to O_2 , producing H_2O . In photosynthesis, H_2O is the source of electrons, which are energized by light, temporarily stored in NADPH, and used to reduce CO_2 to carbohydrate. **10.3** The action spectrum of photosynthesis shows that some wavelengths of light that are not absorbed by chlorophyll *a* are still effective at promoting photosynthesis. The light-harvesting complexes of photosystems contain accessory pigments such as chlorophyll *b* and carotenoids, which absorb different wavelengths and pass the energy to chlorophyll *a*, broadening the spectrum of light usable for photosynthesis.

10.4



In the reduction phase of the Calvin cycle, ATP phosphorylates a three-carbon compound, and NADPH then reduces this compound to G3P. ATP is also used in the regeneration phase, when five molecules of G3P are converted to three molecules of the five-carbon compound RuBP. Rubisco catalyzes the first step of carbon fixation—the addition of CO_2 to RuBP.

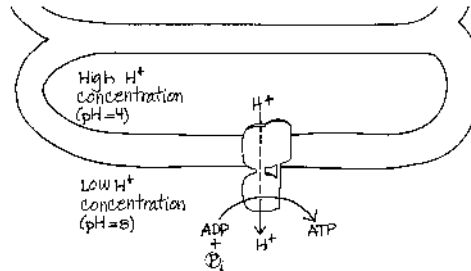
10.5 Both C_4 photosynthesis and CAM photosynthesis involve initial fixation of CO_2 to produce a four-carbon compound (in mesophyll cells in C_4 plants and at night in CAM plants). These compounds are then broken down to release CO_2 (in the bundle-sheath cells in C_4 plants and during the day in CAM plants). ATP is required for recycling the molecule that is used initially to combine with CO_2 . These pathways avoid the photorespiration that consumes ATP

and reduces the photosynthetic output of C_3 plants when they close stomata on hot, dry, bright days. Thus, hot, arid climates would favor C_4 and CAM plants. **10.6** Sucrose made in the leaves of plants is transported through veins to nonphotosynthetic parts of the plant, where some of it is oxidized by cellular respiration, producing ATP for cellular processes. Other sugar molecules enter anabolic pathways, where they are used for synthesis of proteins, lipids, and polysaccharides such as cellulose, the main component of cell walls. Excess sugar is stockpiled as glucose subunits of the polysaccharide starch.

Test Your Understanding

1. D 2. B 3. C 4. A 5. A 6. B 7. C

10.



The ATP would end up outside the thylakoid. The thylakoids were able to make ATP in the dark because the researchers set up an artificial proton concentration gradient across the thylakoid membrane; thus, the light reactions were not necessary to establish the H^+ gradient required for ATP synthesis by ATP synthase.

Chapter 11

Figure Questions

Figure 11.6 Epinephrine is a signaling molecule outside the cell; presumably, it binds to a cell-surface receptor protein and thus is part of the signal reception step. **Figure 11.8** This is an example of passive transport. The ion is moving down its concentration gradient, and no energy is required. **Figure 11.9** The aldosterone molecule, a steroid, doesn't need a receptor protein—it is hydrophobic and can therefore pass directly through the hydrophobic lipid bilayer of the plasma membrane into the cell. (Hydrophilic molecules cannot do this.) **Figure 11.10** The entire phosphorylation cascade wouldn't operate. Regardless of whether or not the signaling molecule was bound, protein kinase 2 would always be inactive and would not be able to activate the purple-colored protein leading to the cellular response. **Figure 11.11** The signaling molecule (cAMP) would remain in its active form and would continue to signal; the pathway would remain active even in the absence of ligand because the cAMP would persist. **Figure 11.12**

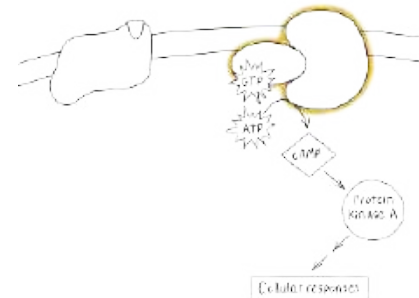


Figure 11.14 The Ca^{2+} pump shown in Figure 11.13 is carrying out active transport of Ca^{2+} ions against their concentration gradient, using ATP for energy. These pumps help maintain a significantly lower $[\text{Ca}^{2+}]$ in the cytoplasm than outside the cell and inside the ER. The Ca^{2+} channel proteins seen in Figure 11.14 are facilitating diffusion of Ca^{2+} down its concentration gradient; here, the Ca^{2+} is moving from inside the ER, where it is more concentrated, to the cytoplasm. Because the ion is moving down its concentration gradient, no energy is required. **Figure 11.16** 100,000,000 (one hundred million, or 10^8) glucose molecules are released just from one epinephrine binding to the receptor. The first step results in $100\times$ amplification (one epinephrine activates 100 G proteins); the next step does not amplify the response; the next step is a $100\times$ amplification (10^2 active adenylyl cyclase molecules to 10^4 cyclic AMPs); the next step does not amplify; the next two steps are each $10\times$ amplifications, and the final step is a $100\times$ amplification. **Figure 11.17** The signaling pathway shown in Figure 11.14 leads to the splitting of PIP_2 into the second messengers DAG and IP_3 , which produce different responses. (The response elicited by DAG is mentioned but not shown.) The pathway shown for cell B is similar in that it branches and leads to two responses.

Concept Check 11.1

1. The two cells of opposite mating type (**a** and **α**) each secrete a unique signaling molecule, which can only be bound by receptors carried on cells of the opposite mating type. Thus, the **a** mating factor cannot bind to another **a** cell and cause it to grow toward the first **a** cell. Only an **α** cell can “receive” the signaling molecule and respond by directed growth.
2. Glycogen phosphorylase acts in the third stage, the cellular response to epinephrine signaling.
3. Glucose 1-phosphate would not be generated because the activation of the